

MIS 443 – FINAL EXAMINATION

Finance Analysis with PostgreSQL

Course	MIS 443	Business Data Management
Duration	90 minutes	Maximum mark: 100

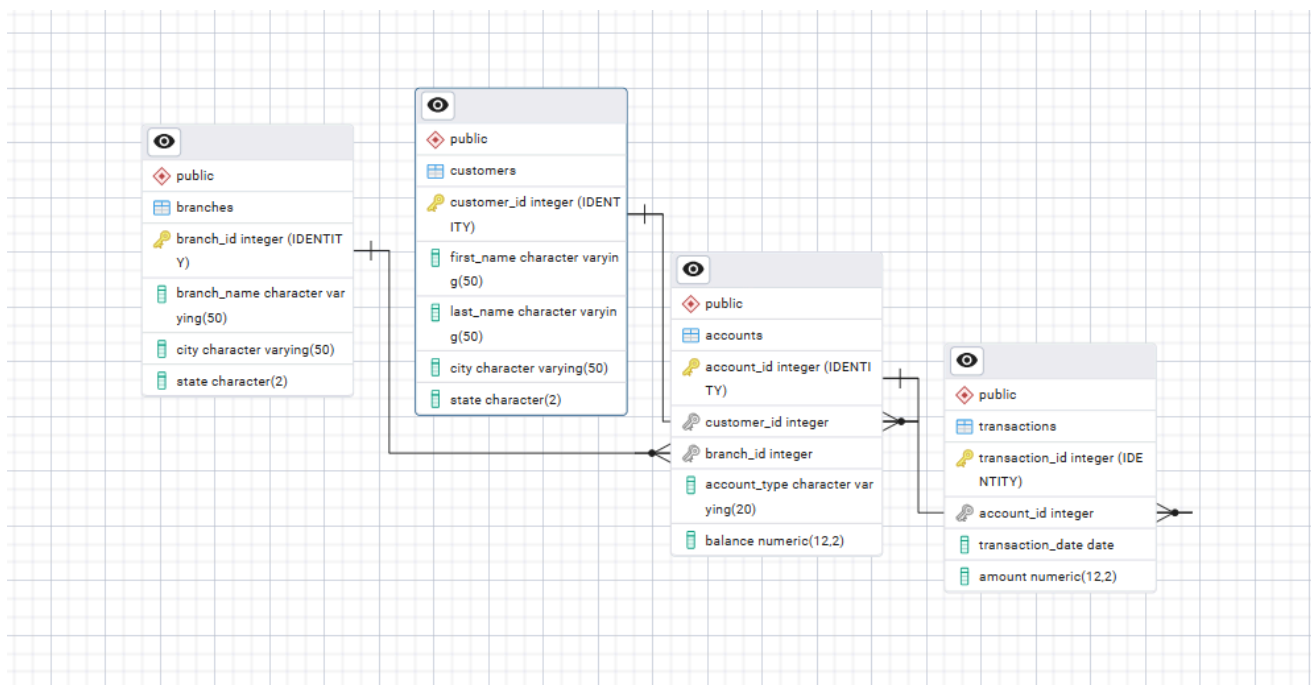
Dataset Introduction

The dataset represents a retail bank that manages customers, branches, bank accounts, and financial transactions. Managers use the data to analyse customer portfolios, account balances, branch performance, and transaction activity.

- Customers: stores customer names and locations.
- Branches: stores branch names and locations.
- Accounts: stores account ownership, branch assignment, account type, and current balance.
- Transactions: stores the date and amount of each account transaction.

Each account belongs to one customer and one branch, and each account may have multiple transactions. Positive account balances represent funds held by customers; negative Credit Card balances represent amounts owed. Positive transaction amounts are credits, while negative amounts are debits.

Entity–Relationship Diagram



Examination Questions

Question 1 – Database Setup (10 marks)

Create a PostgreSQL database using your full name in lowercase, without spaces or Vietnamese diacritics. Connect to the database and execute MIS443_Finance_PostgreSQL.sql. Confirm that the customers, branches, accounts, and transactions tables are available in the public schema.

The screenshot shows the pgAdmin 4 interface. The left pane displays the Object Explorer with the 'public' schema selected. The central pane shows the SQL editor with the following query:

```

32
33 /*
34 QUESTION 1 - DATABASE SETUP (10 marks)
35
36 Create a PostgreSQL database using your full name in lowercase, without spaces
37 or Vietnamese diacritics. Connect to it and execute
38 MIS443_Finance_PostgreSQL.sql. Confirm that customers, branches, accounts, and
39 transactions are available in the public schema. (10 marks)
40 */
41 SELECT 'accounts' AS table_name, COUNT(*) AS row_count FROM public.accounts
42 UNION ALL
43 SELECT 'branches', COUNT(*) FROM public.branches
44 UNION ALL
45 SELECT 'customers', COUNT(*) FROM public.customers
46 UNION ALL
47 SELECT 'transactions', COUNT(*) FROM public.transactions
48 ORDER BY table_name;
49 -- Expected result after loading the supplied data:
50 -- accounts | 15
51 -- branches | 15
52 -- customers | 6
53 -- transactions | 15
54
55

```

The bottom pane shows the Data Output table:

table_name	row_count
accounts	15
branches	15
customers	6
transactions	15

A status bar at the bottom indicates: "Successfully run. Total query runtime: 48 msec. 4 rows affected."

Question 2 – Customer and Account Overview (10 marks)

(a) The Customer Service Manager needs a contact list of customers living in New York. Display customer_id, the customer's full name as full_name, and city. Sort by customer_id. (5 marks)

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure for 'tangsohan' on 'PostgreSQL 18'. The 'customers' table is selected. The main pane shows a SQL query window with the following content:

```

56 /*
57 QUESTION 2 - CUSTOMER AND ACCOUNT OVERVIEW (10 marks)
58
59 (a) The Customer Service Manager needs a contact list of customers living in
60 New York. Display customer_id, full_name, and city, sorted by customer_id.
61 (5 marks)
62 */
63
64 -- Your answer here
65 select customer_id, first_name || ' ' || last_name as full_name, city
66 from customers
67 where city = 'New York'
68 order by customer_id;
69
70 -- Expected result:
71 -- 1 | John Doe | New York
72 -- 2 | Jane Doe | New York
73
74 */
75
76 (b) Management needs to confirm the size of the account portfolio. Calculate
77 the total number of accounts and name the result total_accounts. (5 marks)
78 */
79

```

The Data Output pane shows the results of the first query:

customer_id [PK] integer	full_name text	city character varying (50)
1	John Doe	New York
2	Jane Doe	New York

The status bar at the bottom indicates 'Total rows: 2' and 'Query complete 00:00:00.055'.

(b) Management needs to confirm the size of the account portfolio. Calculate the total number of accounts and name the result `total_accounts`. (5 marks)

The screenshot shows the pgAdmin 4 interface. The main pane shows a SQL query window with the following content:

```

68 order by customer_id;
69
70 -- Expected result:
71 -- 1 | John Doe | New York
72 -- 2 | Jane Doe | New York
73
74 */
75
76 (b) Management needs to confirm the size of the account portfolio. Calculate
77 the total number of accounts and name the result total_accounts. (5 marks)
78 */
79
80 -- Your answer here
81 select count(*) as total_accounts
82 from accounts;
83
84 -- Expected result: 15
85
86 */
87
88 QUESTION 3 - ACCOUNT BALANCE ANALYSIS (20 marks)
89
90 (a) The Finance Manager wants to monitor funds held in checking accounts.
91 Calculate their total balance and name the result total_checking_balance.
92 (10 marks)
93

```

The Data Output pane shows the results of the second query:

total_accounts bigint
15

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.051'. A green message box at the bottom right says 'Successfully run. Total query runtime: 51 msec. 1 rows affected.'

Question 3 – Account Balance Analysis (20 marks)

(a) The Finance Manager wants to monitor funds held in Checking accounts. Calculate their total balance and name the result `total_checking_balance`. (10 marks)

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure for 'tangsohan' (PostgreSQL 18), including schemas like 'public' and tables like 'accounts', 'branches', 'customers', and 'transactions'. The main pane shows a SQL query window with the following content:

```

-- Your answer here
81 select count(*) as total_accounts
82 from accounts;
83 -- Expected result: 15
84
85 /*
86 QUESTION 3 -- ACCOUNT BALANCE ANALYSIS (20 marks)
87
88 (a) The Finance Manager wants to monitor funds held in checking accounts.
89 Calculate their total balance and name the result total_checking_balance.
90 (10 marks)
91 */
92
93 -- Your answer here
94 select sum(balance) as total_checking_balance
95 from accounts
96 where account_type = 'checking';
97 -- Expected result: 31000.00
98
99
100 /*
101 (b) The Los Angeles Regional Manager wants to compare customer portfolios.
102 */

```

The 'Data Output' pane shows the result of the query:

totalChecking_balance
31000.00

Total rows: 1 Query complete 00:00:00.047

(b) For each customer living in Los Angeles, display customer_id, full_name, and total_balance across all account types. Sort by total_balance from highest to lowest. (10 marks)

The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure for 'tangsohan' (PostgreSQL 18). The main pane shows a SQL query window with the following content:

```

103 For each customer living in Los Angeles, display customer_id, full_name, and
104 total_balance across all account types. Sort by total_balance descending.
105 (10 marks)
106 */
107
108 -- Your answer here
109 select c.customer_id, c.first_name || ' ' || c.last_name as full_name, sum(a.balance) as total_balance
110 from customers c
111 join accounts a ON a.customer_id = c.customer_id
112 WHERE CITY = 'Los Angeles'
113 GROUP BY c.customer_id, c.first_name, c.last_name
114 order by total_balance DESC;
115 -- Expected result:
116 -- 5 | Michael Lee | 60000.00
117 -- 6 | Jennifer Wang | 15000.00
118
119
120 /*
121 QUESTION 4 -- BRANCH AND CUSTOMER PORTFOLIO ANALYSIS (20 marks)
122
123 (a) Senior management wants to identify the branch with the highest average
124 account balance. Display branch_id, branch_name, city, and average_balance.
125 Return all branches tied for the highest average and round to two decimals.
126 (10 marks)

```

The 'Data Output' pane shows the result of the query:

customer_id	full_name	total_balance
5	Michael Lee	60000.00
6	Jennifer Wang	15000.00

Total rows: 2 Query complete 00:00:00.072

Successfully run. Total query runtime: 72 msec. 2 rows affected.

Question 4 – Branch and Customer Portfolio Analysis (20 marks)

(a) Identify the branch with the highest average account balance. Display branch_id, branch_name, city, and average_balance. Round to two decimal places and include ties. (10 marks)

pgAdmin 4
File Object Tools Edit View Window Help

Object Explorer

Servers (2)
PostgreSQL 17
PostgreSQL 18
Databases (5)
SCHOOL
postgres
sqlda
tangsohan
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (4)
accounts
branches
customers
transactions
Trigger Functions
Types
Views
Subscriptions
thu

MIS443_Finance_Final_Exam_Skeleton.sql X

tangsohan/postgres@PostgreSQL 18

Query Query History

```

118
119
120 /*
121 QUESTION 4 - BRANCH AND CUSTOMER PORTFOLIO ANALYSIS (20 marks)
122
123 (a) Senior management wants to identify the branch with the highest average
124 account balance. Display branch_id, branch_name, city, and average_balance.
125 Return all branches tied for the highest average and round to two decimals.
126 (10 marks)
127 */
128 select b.branch_id, b.branch_name, b.city, round(avg(a.balance),2) as average_balance
129 from branches b
130 join accounts a on a.branch_id = b.branch_id
131 group by b.branch_id, b.branch_name, b.city
132 having avg(a.balance) = (
133   SELECT MAX(avg_bal)
134   from (
135     select avg(balance) as avg_bal
136     from accounts
137     group by branch_id
138   ) t
139 );
140
141 -- Your answer here

```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

branch_id	branch_name	city	average_balance
[PK] integer	character varying (50)	character varying (50)	numeric
14	North Beach	San Francisco	30000.00

Total rows: 1 Query complete 00:00:00.056

LF Ln 141, Col 20 11:04 AM 06-Aug-26

(b) Identify the customer who owns the single account with the highest current balance. Display customer_id, full_name, account_id, account_type, and balance. Include ties. (10 marks)

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Object Explorer

Servers (2)
PostgreSQL 17
PostgreSQL 18
Databases (5)
SCHOOL
postgres
sqlda
tangsohan
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (4)
accounts
branches
customers
transactions
Trigger Functions
Types
Views
Subscriptions
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tangsohan/postgres@PostgreSQL 18

Query Query History

```

142
143 -- Expected result:
144 -- 14 | North Beach | San Francisco | 30000.00
145
146
147 /*
148 (b) A relationship manager wants to identify the customer who owns the single
149 account with the highest current balance. Display customer_id, full_name,
150 account_id, account_type, and balance. Include ties if any. (10 marks)
151 */
152
153 -- Your answer here
154 select c.customer_id,
155        c.first_name || ' ' || c.last_name AS full_name,
156        a.account_id,
157        a.account_type,
158        a.balance
159 from customers c
160 join accounts a on c.customer_id = a.customer_id
161 where a.balance = (SELECT MAX(balance) FROM accounts);
162
163 -- Expected result:
164 -- 5 | Michael Lee | 10 | Savings | 50000.00
165
166 -- Your answer here

```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

customer_id	full_name	account_id	account_type	balance
integer	text	integer	character varying (20)	numeric (12,2)
5	Michael Lee	10	Savings	50000.00

Total rows: 1 Query complete 00:00:00.046

LF Ln 161, Col 56 11:05 AM 06-Aug-26

Question 5 – Customer Value and Activity (20 marks)

(a) Identify the most active customers based on the total number of transactions across all their accounts. Display customer_id, full_name, and total_transactions. Include ties. (10 marks)

pgAdmin 4
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Object Explorer

Servers (2)
PostgreSQL 17
PostgreSQL 18
Databases (5)
SCHOOL
postgres
sqlda
tangsohan
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (4)
accounts
branches
customers
transactions
Trigger Functions
Types
Views
Subscriptions
thu

MIS443_Finance_Final_Exam_Skeleton.sql X

tangsohan/postgres@PostgreSQL 18

Query Query History

```

1677
1678 (a) The Customer Relationship Manager wants to identify the most active
1679 customer based on the total number of transactions across all their accounts.
1680 Display customer_id, full_name, and total_transactions. Include ties.
1681 (10 marks)
1682 */
1683
1684 -- Your answer here
1685
1686 SELECT c.customer_id,
1687        c.first_name || ' ' || c.last_name AS full_name,
1688        COUNT(t.transaction_id) AS total_transactions
1689 FROM   customers c
1690 JOIN   accounts a ON c.customer_id = a.customer_id
1691 JOIN   transactions t ON a.account_id = t.account_id
1692 GROUP BY c.customer_id, c.first_name, c.last_name
1693 HAVING COUNT(t.transaction_id) = (
1694     SELECT MAX(cnt)
1695     FROM (
1696         SELECT COUNT(t.transaction_id) AS cnt
1697         FROM   accounts a2
1698         JOIN   transactions t2 ON a2.account_id = t2.account_id
1699         GROUP BY a2.customer_id
1700     ) s
1701 );

```

Data Output Messages Notifications

customer_id [PK] integer	full_name text	total_transactions bigint
1	4 Alice Johnson	4
2	2 Jane Doe	4

Total rows: 2 Query complete 00:00:00.072 LF Ln 190, Col 3

11:07 AM 06-Aug-26

(b) Identify the customer with the highest combined balance across Checking and Savings accounts. Exclude Credit Card accounts. Display customer_id, full_name, and total_deposit_balance. Include ties. (10 marks)

pgAdmin 4
File Object Tools Edit View Window Help

Object Explorer

Servers (2)
PostgreSQL 17
PostgreSQL 18
Databases (5)
SCHOOL
postgres
sqlda
tangsohan
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas (1)
public
Aggregates
Collations
Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (4)
accounts
branches
customers
transactions
Trigger Functions
Types
Views
Subscriptions
thu

MIS443_Finance_Final_Exam_Skeleton.sql X

tangsohan/postgres@PostgreSQL 18

Query Query History

```

1706
1707 (b) The Deposit Manager wants to identify the customer with the highest total
1708 balance across Checking and Savings accounts only. Display customer_id,
1709 full_name, and total_deposit_balance. Include ties. (10 marks)
1710 */
1711
1712 -- Your answer here
1713
1714 SELECT c.customer_id,
1715        c.first_name || ' ' || c.last_name AS full_name,
1716        SUM(a.balance) AS total_deposit_balance
1717 FROM   customers c
1718 JOIN   accounts a ON c.customer_id = a.customer_id
1719 WHERE  a.account_type IN ('Checking', 'Savings')
1720 GROUP BY c.customer_id, c.first_name, c.last_name
1721 HAVING SUM(a.balance) = (
1722     SELECT MAX(tot)
1723     FROM (
1724         SELECT SUM(balance) AS tot
1725         FROM   accounts
1726         WHERE  account_type IN ('Checking', 'Savings')
1727         GROUP BY customer_id
1728     ) s
1729 );
1730 -- Expected result:

```

Data Output Messages Notifications

customer_id [PK] integer	full_name text	total_deposit_balance numeric
1	5 Michael Lee	60000.00

Total rows: 1 Query complete 00:00:00.055 LF Ln 211, Col 20

11:07 AM 06-Aug-26

Question 6 – Advanced Finance Analysis (20 marks)

(a) Identify the branch with the highest total balance across all account types. Display branch_id, branch_name, and total_balance. Include ties. (10 marks)

pgAdmin 4
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Object Explorer

- Servers (2)
 - PostgreSQL 17
 - PostgreSQL 18
 - Databases (5)
 - SCHOOL
 - postgres
 - sqlda
 - tangsohan
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (4)
 - accounts
 - branches
 - customers
 - transactions
 - Trigger Functions
 - Types
 - Views
 - Subscriptions

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Type here to search

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tangsohan/postgres@PostgreSQL 18

Query Query History

224 QUESTION 6 - ADVANCED FINANCE ANALYSIS (20 marks)

225

226 (a) Management wants to identify the branch with the highest total balance

227 across all account types. Display branch_id, branch_name, and total_balance.

228 Include ties. (10 marks)

229 */

230

231 -- Your answer here

232 SELECT b.branch_id,

233 b.branch_name,

234 SUM(a.balance) AS total_balance

235 FROM branches b

236 JOIN accounts a ON b.branch_id = a.branch_id

237 GROUP BY b.branch_id, b.branch_name

238 HAVING SUM(a.balance) = (

239 SELECT MAX(tot)

240 FROM (

241 SELECT SUM(balance) AS tot

242 FROM accounts

243 GROUP BY branch_id

244) s

245);

246 -- Expected result:

247 -- 14 | North Beach | 60000.00

248

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

branch_id	branch_name	total_balance
[PK] integer	character varying (50)	numeric
14	North Beach	60000.00

Total rows: 1 Query complete 00:00:00.052 LF Ln 247, Col 25

11:00 AM 06-Aug-26

(b) Rank all customers by total balance across all account types. Equal totals must receive the same rank without gaps. Display customer_id, full_name, total_balance, and balance_rank. Do not use a CTE. (5 marks)

pgAdmin 4
File Object Tools Edit View Window Help

Object Explorer

- Servers (2)
 - PostgreSQL 17
 - PostgreSQL 18
 - Databases (5)
 - SCHOOL
 - postgres
 - sqlda
 - tangsohan
 - Cast
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas (1)
 - public
 - Aggregates
 - Collations
 - Domains
 - FTS Configurations
 - FTS Dictionaries
 - FTS Parsers
 - FTS Templates
 - Foreign Tables
 - Functions
 - Materialized Views
 - Operators
 - Procedures
 - Sequences
 - Tables (4)
 - accounts
 - branches
 - customers
 - transactions
 - Trigger Functions
 - Types
 - Views
 - Subscriptions

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MIS443_Finance_Final_Exam_Skeleton.sql X

tangsohan/postgres@PostgreSQL 18

Query Query History

248

249

250 /*

251 (b) Rank all customers by total balance across all account types. Equal totals

252 must receive the same rank without gaps. Display customer_id, full_name,

253 total_balance, and balance_rank. Do not use a CTE. (5 marks)

254 */

255

256 -- Your answer here

257 SELECT c.customer_id,

258 c.first_name || ' ' || c.last_name AS full_name,

259 SUM(a.balance) AS total_balance,

260 RANK() OVER (ORDER BY SUM(a.balance) DESC) AS balance_rank

261 FROM customers c

262 JOIN accounts a ON c.customer_id = a.customer_id

263 GROUP BY c.customer_id, c.first_name, c.last_name

264 ORDER BY balance_rank;

265 -- Expected result:

266 -- 5 | Michael Lee | 60000.00 | 1

267 -- 4 | Alice Johnson | 25000.00 | 2

268 -- 3 | Bob Smith | 20500.00 | 3

269 -- 6 | Jennifer Wang | 15000.00 | 4

270 -- 2 | Jane Doe | 11500.00 | 5

271 -- 1 | John Doe | 5500.00 | 6

Data Output Messages Notifications

Showing rows: 1 to 6 Page No: 1 of 1

customer_id	full_name	total_balance	balance_rank
[PK] integer	text	numeric	bigint
5	Michael Lee	60000.00	1
4	Alice Johnson	25000.00	2
3	Bob Smith	20500.00	3
6	Jennifer Wang	15000.00	4
2	Jane Doe	11500.00	5
1	John Doe	5500.00	6

Total rows: 6 Query complete 00:00:00.059

Successfully run. Total query runtime: 59 msec. 6 rows affected.

LF Ln 254, Col 24

11:00 AM 06-Aug-26

(c) Use a CTE to calculate the total number of transactions for every branch, including branches with no transactions. Return the branch or branches with the highest total. Display branch_id, branch_name, and total_transactions. (5 marks)

The screenshot shows the pgAdmin 4 interface with a PostgreSQL database named 'tangsohan'. The 'Query' tab is active, displaying a SQL query that uses a CTE to calculate the total number of transactions for every branch, including branches with no transactions. The query is as follows:

```

273
274 /*
275 (c) Use a CTE to calculate the total number of transactions for every branch,
276 including branches with no transactions. Return the branch or branches with
277 the highest total. Display branch_id, branch_name, and total_transactions.
278 (5 marks)
279 */
280
281 -- Your answer here
282 WITH branch_tx AS (
283   SELECT b.branch_id,
284          b.branch_name,
285          COUNT(t.transaction_id) AS total_transactions
286   FROM   branches b
287   LEFT JOIN accounts a ON b.branch_id = a.branch_id
288   LEFT JOIN transactions t ON a.account_id = t.account_id
289   GROUP BY b.branch_id, b.branch_name
290 )
291 SELECT branch_id, branch_name, total_transactions
292 FROM   branch_tx
293 WHERE  total_transactions = (SELECT MAX(total_transactions) FROM branch_tx);
294
295 -- Expected result:
296 -- 1 | Main | 4
297 -- 8 | South Bay | 4

```

The 'Data Output' tab shows the results of the query, displaying two rows of data:

branch_id	branch_name	total_transactions
1	Main	4
8	South Bay	4

The status bar at the bottom indicates 'Total rows: 2' and 'Query complete 00:00:00.059'.

Mark Distribution

Question	Marks
Question 1	10
Question 2	10
Question 3	20
Question 4	20
Question 5	20
Question 6	20

END OF EXAMINATION